

The Compute Requirements of Embodied Autonomy

A tier-by-tier arithmetic and precision analysis of the canonical autonomy pipeline

Claims about how much compute an autonomous platform needs are usually asserted. This annex derives the number. It decomposes the autonomy pipeline into seven tiers, identifies the core algorithms in each, estimates sustained arithmetic demand and the numeric precision each tier actually requires, and allocates a latency budget for two operating regimes. It then converts that demand into a silicon requirement using the measured efficiency data in the companion annex.

This annex states the **argument**. Its companion, the Workload Data Annex, states the **inputs**: every platform parameter, the op-count derivation for each of the seventeen stages, the ingress/transformation/egress model behind each op count, the update rate assumed in each regime with the reason for it, and the latency allocation each stage is given. Nothing is asserted here that is not derived there. A reader who accepts the reasoning needs only this document; a reader who wants to challenge a number needs the other one.

The analysis is deliberately **architecture-generic**. At least a dozen published research and commercial autonomy stacks instantiate the same seven tiers with different component choices; the conclusions here depend on the *structure* of the pipeline, not on any one implementation. Where a concrete sensor suite and compute budget are needed to make the arithmetic specific, this annex uses a published open reference design — an Orin NX-class mission computer with a 25 W compute allocation, dual 1440×1080 grayscale cameras, a color camera, a 360°×90° 30 m LiDAR, a 180°×180° 49 m radar, and an 800 Hz IMU. Substituting a comparable suite changes individual line items but not the aggregate conclusion.

1 • THE CANONICAL PIPELINE

Every fielded autonomy stack the authors are aware of decomposes into the same seven tiers. The names differ; the dependency structure does not.

- **T1 Sensor front-end.** Deskew, rectification, disparity, FFT-domain radar processing. Converts raw transducer output into geometrically consistent measurements.
- **T2 State estimation.** Visual/LiDAR-inertial odometry and factor-graph smoothing. Answers “where am I,” at high rate, without GNSS.
- **T3 Dense mapping.** Volumetric occupancy or truncated/Euclidean signed-distance fields. Answers “what is around me and how far away is it.”
- **T4 Semantic perception.** Detection, segmentation, open-vocabulary grounding, and vision-language reasoning. Answers “what is it and does it matter.”
- **T5 Deliberative planning.** Sampling-based graph construction, information-gain evaluation, global routing. Answers “where should I go.”
- **T6 Reactive navigation and safety.** Learned policies, model-predictive control, and barrier-function safety filters. Answers “what do I do in the next 10 milliseconds.”
- **T7 Low-level control.** Attitude and rate loops on the flight controller.

2 · STAGE-BY-STAGE ARITHMETIC AND PRECISION

Sustained demand at the sizing regime — **air superiority**, the heaviest of the five. Op counts are structural estimates derived from published algorithm definitions and the sensor configuration above; they are engineering estimates, not measurements. The numeric floor is the *lowest* precision at which the stage remains numerically sound in production implementations; a mixed stage is bound by its hardest component, not its dominant one. The op-per-byte column is arithmetic intensity, and it is the column that explains the measured efficiency: **12 of the 17 stages move more than one byte for every fifteen operations.**

Stage	Core algorithms	Ops/s at sizing regime	Op per byte	Numeric floor	Latency budget min / sizing
T1 · Sensor front-end					
LiDAR deskew + voxel downsample	SE(3) interpolation per point; spatial hash	0.21 GOP/s	1.0	FP32/64	5 / 2 ms
Stereo rectify + SGM disparity	Semi-global matching, 128 disparities	238.9 GOP/s	4.9	INT8 + FP32	25 / 4 ms
Radar FMCW post-processing	Range/Doppler FFT, CFAR, angle estimate	20.0 GOP/s	11.9	INT8 + FP32	10 / 3 ms
T2 · State estimation					
LiDAR-inertial odometry (factor graph)	ikd-tree kNN, point-to-plane Huber, IMU preintegration, windowed Gauss-Newton	8.0 GOP/s	3.9	FP32/64	40 / 8 ms
Visual front-end (2 cameras)	FAST/Shi-Tomasi extraction + pyramidal KLT tracking	13.2 GOP/s	13.3	INT8 + FP32	20 / 4 ms
Windowed bundle adjustment	Schur complement, Levenberg-Marquardt	10.5 GOP/s	6.3	FP32/64	60 / 15 ms
T3 · Dense mapping					
TSDF integration (volumetric map)	Bundled raycasting, weighted truncated SDF update	3.9 GOP/s	1.3	FP32/64	50 / 40 ms
ESDF propagation	Incremental wavefront / brushfire over the TSDF	8.0 GOP/s	7.7	FP32/64	50 / 40 ms
T4 · Semantic perception					
Open-vocabulary detection	YOLOe-class detector + cached text embeddings	9.0 TOP/s	500	INT8 + FP16	200 / 16 ms
Onboard vision-language reasoning	2B-parameter VLM: 600-token prefill + 64-token decode	0.00 GOP/s	20.4	INT8 + FP16	0 / — ms
T5 · Deliberative planning					
Volumetric gain evaluation	Frustum raycasting per candidate vertex over unknown space	0.00 GOP/s	1.0	FP32/64	500 / — ms
Graph construction + collision check	Sampling-based graph, edge collision queries, Dijkstra	0.25 GOP/s	7.8	FP32/64	300 / 400 ms
T6 · Reactive navigation & safety					
Learned SDF encoder (neural SDF-MPC)	Convolutional range-image encoder + MLP SDF head	2.5 TOP/s	161	INT8 + FP16	20 / 2 ms
Learned navigation policy (DRL)	Depth collision encoder + actor MLP	1.0 TOP/s	143	INT8 + FP16	20 / 2 ms
Nonlinear MPC solve	SQP / interior point over a 30-step horizon	2.5 GOP/s	7.2	FP32/64	10 / 1.5 ms
Composite CBF safety filter	Weighted virtual-obstacle QP over a sliding map window	0.50 GOP/s	3.8	FP32/64	2 / 0.5 ms
T7 · Low-level control					
Attitude / rate control	Cascaded PID or geometric controller on the autopilot	0.40 GOP/s	200	FP32/64	2 / 0.5 ms

3 • WHY THREE PRECISION CLASSES EXIST

The single most consequential fact about this workload is that it is not one numeric regime. It is three, and only the first is served by the INT8 tensor path that embedded SoC datasheets advertise.

Class A — INT8-eligible

Dense convolution and GEMM in feature-extraction backbones: detector and segmentation trunks, depth and range-image encoders, the convolutional stages of learned navigation policies. Weights and activations in these layers have bounded, well-behaved distributions, and post-training quantization with per-channel scaling is routine. This is the work the tensor cores were built for.

Class B — FP16 floor

Softmax, layer and batch normalization, GELU/SiLU activations, attention score computation, and regression heads that emit signed or unbounded quantities. These operations either require exponentials over a wide dynamic range or accumulate variance across a tensor; INT8 representations distort the ranking of detection confidences and destroy the numerical behavior of normalization. In practice every production INT8 deployment keeps these layers in FP16 — the network is quantized, the *pipeline* is mixed. For transformer-based vision-language components the attention path is FP16 or, at best, FP8 with per-tensor scaling.

Class C — FP32/FP64 floor

Everything geometric or constrained. Factor-graph state estimation propagates covariances and forms normal-equation Hessians whose condition numbers routinely reach 10^6 – 10^{10} ; FP16 carries eleven mantissa bits and saturates at 65,504, so a single $J^T J$ entry overflows. Signed-distance fields need fine resolution near surfaces and wide range far from them simultaneously. Model-predictive control and barrier-function filters solve constrained optimization problems where a quantization error does not degrade an estimate — it violates a constraint, and a violated safety constraint is a collision. Production implementations of all of these use FP64.

This classification is the mechanism behind the measurement in the companion annex. A datasheet TOPS figure describes Class A throughput. An autonomy pipeline spends most of its schedule outside Class A.

4 · AGGREGATE DEMAND ACROSS FIVE OPERATING REGIMES

Compute demand is not one number for a class of platform. It is a function of the regime, because different regimes are governed by different physics and load different tiers. **Minimum usable** is low-speed inspection with semantics offloaded to a datalink. **Contested** moves vision-language reasoning onboard because a jammed platform has no alternative. **Far flight** is energy-governed endurance ISR, where wide-area search and route optimization dominate. **Fast flight** is latency-governed dash and small-UAS intercept. **Air superiority** is fighter-class intercept at 150–250 m/s, where no onboard sensor reaches the stopping distance and the problem becomes engagement geometry rather than obstacle avoidance.

Exhibit A — Where the arithmetic goes, by regime

Totals: 1.2 / 5.3 / 10.2 / 7.2 / 12.8 TOP/s. Each regime loads a different part of the pipeline — far flight stresses search and routing; air superiority stresses sensing, guidance and control.

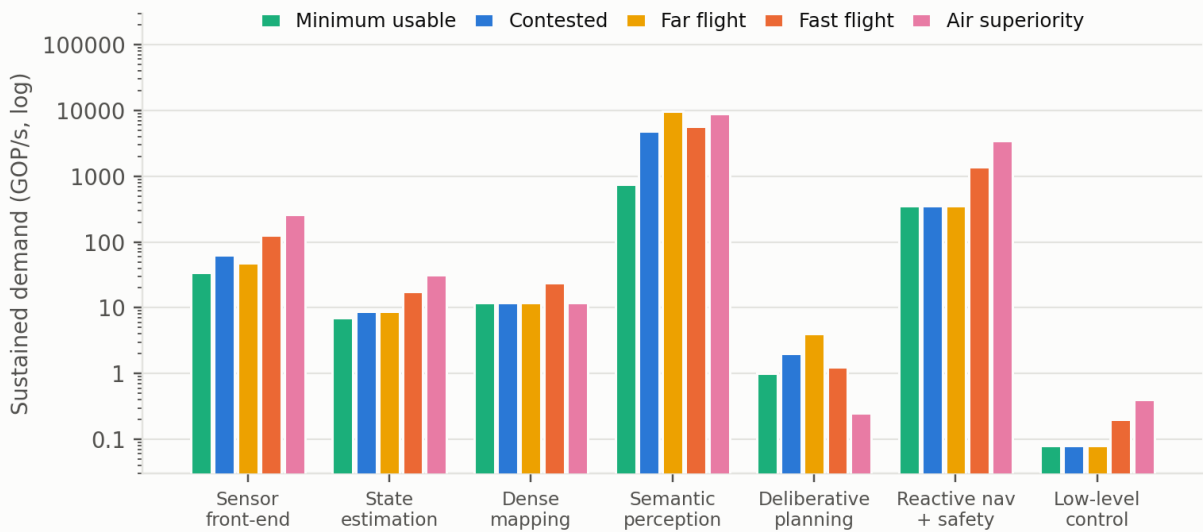


Exhibit A. Per-tier sustained arithmetic demand across all five regimes. No two regimes load the same tier hardest: far flight peaks on semantic perception and deliberative planning, air superiority on the sensor front-end and the reactive control loop while its planning tier collapses to near zero.

Operating regime	Sustained demand	Class A share	Class C share	Governing constraint
Minimum usable	1.15 TOP/s	82.1%	2.1%	Operator in the loop
Contested	5.34 TOP/s	69.4%	0.6%	Emissions control
Far flight	10.24 TOP/s	69.0%	0.3%	Energy — endurance
Fast flight	7.23 TOP/s	72.6%	0.8%	Latency — time to react
Air superiority	12.81 TOP/s	83.4%	0.6%	Engagement geometry

The pipeline sizes on **air superiority at 12.81 TOP/s**. That is the figure carried through the rest of this annex. It is 77% above the fast-flight case and 11× the minimum-usable case, and the spread between regimes is the quantitative argument against a single fixed silicon allocation: sensor front-end demand spans 5.4× between the lightest and heaviest high-performance regime, reactive navigation 10.0×, and deliberative planning 16.3× in the *opposite* direction.

5 • THE SECOND CONSTRAINT: BANDWIDTH, NOT ARITHMETIC

Arithmetic demand is only half the requirement. The companion data annex models the ingress, egress and internal memory traffic of every stage, and the resulting picture is a second constraint that binds independently of the first — and, in three of the five regimes, binds sooner.

The governing quantity is **arithmetic intensity**: operations performed per byte moved. A processor with a 275 TOP/s peak and a 205 GB/s memory interface reaches its peak only on work above roughly 1,340 op per byte. Below that the memory system sets the rate and the multiplier array idles, however large it is. The autonomy pipeline is nowhere near that line:

Stage	Op per byte	Why
LiDAR deskew + voxel downsample	0.98	One pass over the cloud plus a hash probe per point
Volumetric gain evaluation	1.00	96 M scattered voxel queries to accumulate 768 MOP of gain
TSDf integration (volumetric map)	1.25	19.7 M read-modify-writes scattered across the voxel hash
Composite CBF safety filter	3.81	Sweeps a local distance-field window for a 3-variable solve
Attitude / rate control	200	Runs on the flight-control microcontroller, not the mission computer
Open-vocabulary detection	500	Dense convolution — the one stage a tensor array can actually feed

The full table is in the data annex. The summary is that **12 of 17 stages sit below 15 op per byte**, and the single highest in the pipeline is 500. Detection is the only stage whose arithmetic intensity a dense accelerator can exploit — and it is one stage out of seventeen. Nothing in an embodied autonomy stack resembles the dense GEMM that datasheet throughput figures are quoted against.

Multiplying each stage’s per-call traffic by its update rate gives the sustained bandwidth the pipeline demands:

Quantity	Min usable	Contested	Far flight	Fast flight	Air superiority
Sensor ingress (GB/s)	0.36	0.44	0.53	0.89	1.71
Total DRAM traffic (GB/s)	16	156	289	182	100
× a 204.8 GB/s LPDDR5 bus	0.08×	0.76×	1.41×	0.89×	0.49×

Sensor ingress is trivial — under 2 GB/s in the hardest regime. Essentially all of the traffic is generated *inside* the pipeline: cost volumes, voxel hashes, wavefront queues, and autoregressive weight passes. Far flight demands **289 GB/s**, which exceeds the peak bandwidth of an Orin-class bus outright — and peak is not achievable, since sustained bandwidth under scattered access runs 60–70% of it. The effective shortfall is nearer 2×

Two consequences follow, and both matter for silicon selection. First, **the regime ordering of bandwidth is not the regime ordering of arithmetic**. Air superiority demands the most arithmetic of any regime and only half the bus; far flight demands 20% less arithmetic and 2.9× the memory traffic, because wide-area search runs a language model at a sustained rate. Sizing one part for both means paying for bandwidth the interceptor never uses and control-loop throughput the ISR platform never uses. Second, **a single stage sets the ceiling** in three of the five regimes: vision-language decode alone accounts for 260 of far flight’s 289 GB/s, because it re-reads all 2 GB of its weights for each of 64 generated tokens. That stage is not limited by the difficulty of its arithmetic — 4 GOP per token — but by the fact that its weights will not stay on chip.

This is also why quantization is a weaker lever than it appears, and why the measured corpus in the companion annex shows FP16 and BF16 producing near-identical energy efficiency. INT8 halves weight and activation bytes against FP16, but the voxel fields, cost volumes and factor-graph working sets are FP32 or FP64 by requirement rather than by choice. **The bandwidth-dominant stages are precisely the ones quantization cannot touch.**

6 · WHY QUANTIZATION DOES NOT CLOSE THE GAP

At the sizing regime **83% of the arithmetic is INT8-eligible**, which invites the conclusion that aggressive quantization solves the problem. It does not, because arithmetic share and schedule share are not the same quantity. Class A work runs on the tensor path at high utilization. Class B work is elementwise and bandwidth-bound. Class C work is irregular, serial and largely confined to the CPU cluster.

Exhibit B — Quantization cannot reach the work that costs the time

83% of the arithmetic is INT8-eligible, but it consumes only 31% of the schedule at the sizing regime (air superiority). The FP16 and FP32/FP64 tiers own the rest and cannot be quantized.

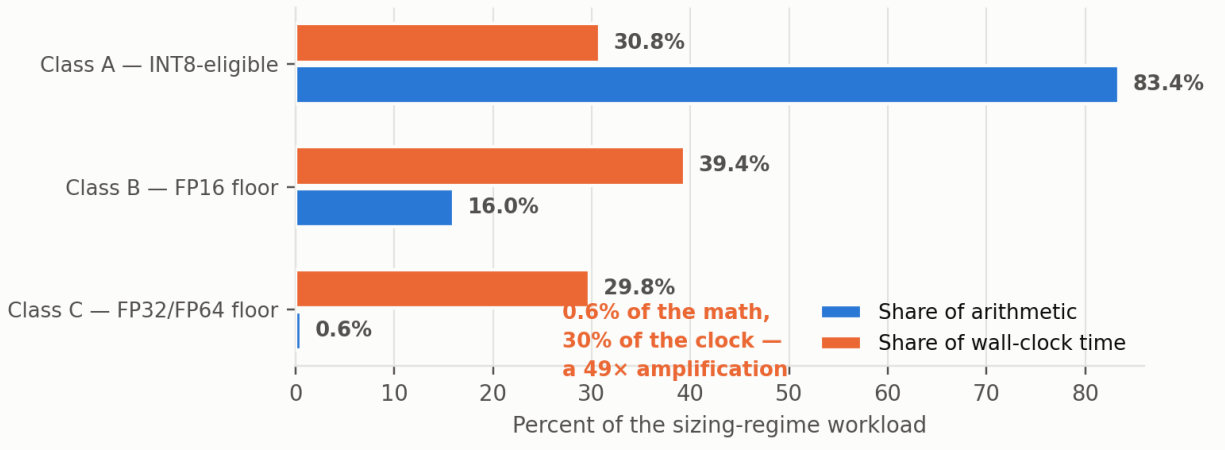


Exhibit B. Arithmetic share against wall-clock share at the sizing regime. Class C is 0.6% of the operations and 30% of the schedule — a 49x amplification. Quantizing Class A perfectly, to zero cost, would remove under a third of the execution time.

The consequence is blunt. Perfect INT8 quantization of every eligible layer — a physically unachievable upper bound — removes **30.8% of the schedule**. The remaining **69.2%** is untouched, because no quantization scheme reaches attention softmax, factor-graph Cholesky, signed-distance fields, or a constrained QP. Class C is the sharpest case: **0.6% of the operations consume 30% of the clock**. An accelerator that leaves that work on a host CPU is capped near **3.4x** at the system level by Amdahl's law, no matter how good its fabric is on the other 99% of the arithmetic.

7 • FROM ARITHMETIC DEMAND TO SILICON REQUIREMENT

The derivation has four steps, each traceable to a measured or published quantity.

Step		Result
1	Compute budget of a representative mission computer (published reference design)	25 W
2	Best energy efficiency measured across 360 configurations on Jetson AGX Orin	101.8 GOPS/W
3	Therefore deliverable inside the envelope, at the measured ceiling (25 W × 101.8)	2.5 TOPS
4	Sustained demand of the pipeline, contested through air-superiority regimes	5.3 – 12.81 TOP/s

The platform is therefore short by **2.1× to 5.1× against the best efficiency ever measured**, and by 175× to 420× against the measured batch-1 median that a real-time control loop actually operates at. Stated as power at the measured ceiling: the sizing regime needs **126 W of compute against a 25 W budget**. That is the gap, in the units the airframe cares about.

Converting the demand into a **nameplate** silicon requirement requires the achieved-to-peak ratio. The companion annex measures batch-1 achieved efficiency between 0.008% and 12% of peak, with a median of 0.31%; a mixed pipeline with a large Class B and Class C fraction realistically sustains 2–5%. Sourcing that work therefore requires:

Achieved-to-peak ratio	Contested	Far flight	Fast flight	Air superiority
5% (optimistic)	107 TOPS	205 TOPS	145 TOPS	256 TOPS
2% (realistic mixed pipeline)	267 TOPS	512 TOPS	361 TOPS	640 TOPS

This is the origin of the ~250 TOPS figure, and the five-regime model shows it to be the **conservative** end of the range rather than the headline. 250 TOPS is roughly what the contested regime needs at a realistic 2% achieved-to-peak, or what air superiority needs at an optimistic 5%. The interceptor case at 2% is **640 TOPS**. None of these figures claims the algorithms perform that many operations per second — they perform between 5 and 12.81 trillion. They are the **nameplate silicon a conventional architecture must carry** in order to deliver that useful work on this mix.

The closing observation is the one that matters for airframe design. At the ~2.3 TOPS/W dense-INT8 peak efficiency of current embedded SoC silicon, **250 nameplate TOPS costs 109 W** and the 640 TOPS interceptor case costs 278 W — against a 25 W budget, on a platform where every watt of compute is a watt subtracted from propulsion and every gram of heatsink from payload. The requirement is not merely unmet. On this architecture it is unmeetable.

8 • ASSUMPTIONS AND SENSITIVITY

- **Op counts and byte counts are structural estimates, not measurements.** Both are derived from published algorithm definitions, standard model architectures, and the stated sensor configuration. Memory-traffic figures additionally assume competent cache blocking; a poorly tiled implementation moves an order of magnitude more, so they are lower bounds on what a fielded system will move.
- **Vision-language reasoning is the largest single term in three regimes.** Removing onboard VLM inference reduces far-flight demand from 10.24 to 4.92 TOP/s and its bandwidth from 289 to 29 GB/s. But a contested or emissions-restricted platform cannot offload it, which is precisely why it is in the requirement. It is absent from the air-superiority regime, where there is no time for deliberative reasoning inside the engagement.
- **Effective throughputs are stated assumptions.** Class A is anchored at 2,000 GOP/s to the fastest batch-1 result in the measured corpus, scaled for INT8 and a smaller part; Class B at 300 and Class C at 15 GOP/s are generous allowances. Tripling the Class B and Class C figures reduces sizing-regime oversubscription from 17.3× to roughly 6× — it does not remove it.
- **Latency figures are budget allocations, not measurements.** They express what each stage must occupy for the loop to close, not what any particular implementation achieves. They are inputs to the model and are listed in full in the data annex.
- **Scheduling and kernel-launch overhead is excluded, and it runs against us.** The MPC and barrier-function stages are serial chains of small dependent kernels. At a few microseconds of launch overhead each, a 300-step SQP recursion spends more time being dispatched than computing. Their real latency is dominated by a term this model does not include.

9 • THE LATENCY CONSEQUENCE, AND WHERE EACH REGIME BREAKS

An oversubscribed pipeline does not announce itself as a failure. Rates degrade, queues lengthen, and glass-to-actuator latency stretches. At the effective throughputs above, the pipeline demands **2.7 / 9.2 / 16.2 / 12.9 / 17.3 seconds of compute per second of flight** across the five regimes. A reactive path designed to close in 80 ms closes in 0.74 s contested and 1.39 s at the sizing regime.

Exhibit C — A compute deficit does not slow the robot; it caps the airframe

Maximum speed at which the platform can still arrest inside its 30 m sensing horizon at 10 m/s².
An oversubscribed pipeline stretches the loop, and the speed ceiling falls with it.

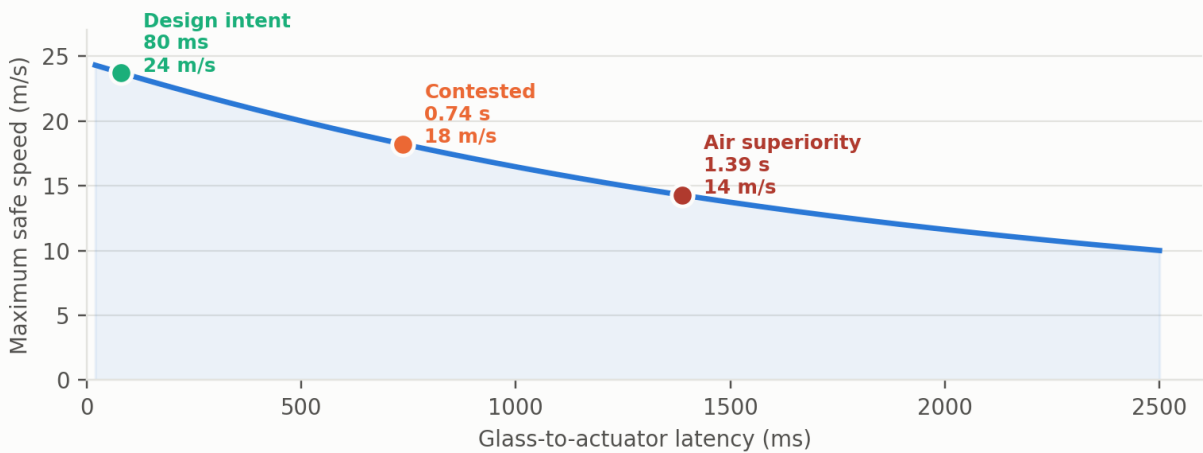


Exhibit C. Maximum speed at which a platform can still arrest inside a 30 m sensing horizon at 10 m/s², as a function of end-to-end latency. Compute deficit converts directly into an airspeed ceiling: 24 m/s at the design-intent loop rate, 14 m/s when the pipeline is oversubscribed 17×.

The aggregate figure conceals the more useful result, which is that **no single stage is the bottleneck across regimes**. Dividing each stage’s service time by the latency its regime allows gives a per-stage constraint ratio; anything at or above 1.00 is a stage that misses its deadline even with the entire accelerator to itself. The data annex plots the full matrix. The counts and the binding tiers are:

Regime	Stages over	Binding tier	What an architect optimizes first
Minimum usable	2 of 16	Sensor front-end, dense mapping	Marginal — both are FP32 stages with poor locality. Essentially servable today.
Contested	4 of 17	Semantic perception	The VLM decode path. Memory-bound, so it is a bandwidth problem, not an arithmetic one.
Far flight	4 of 17	Semantic perception	Search throughput. Loosest latency budgets of the three hard regimes and still the worst single ratio (9.4×), because wide-area search runs the highest sustained detection and reasoning rate.
Fast flight	7 of 17	Dense mapping and the guidance chain	Seven stages miss, spread across four tiers. No single hot spot — this regime demands a different machine, not a faster one.
Air superiority	8 of 15	Sensor front-end and state estimation	Sensing rate and the guidance loop. Deliberative planning runs 25× inside budget here while every sensing and estimation stage misses.

These ratios assume each stage owns the whole machine, so they are lower bounds; applying the oversubscription factors above multiplies them by roughly an order of magnitude. The architectural conclusion does not depend on the magnitude. It depends on the *pattern*: far flight fails on semantic perception while its guidance loop is comfortable; air superiority fails on the sensor front-end and state estimation while its planning tier runs 25× inside budget. A fixed silicon allocation must be sized for the union of every regime’s peak — paying area and power for units idle in any given mission — or accept that it cannot fly whichever regime it was not sized for.

For an interceptor the arithmetic is harsher still, because the relevant quantity is not stopping distance but terminal miss distance. At 200 m/s the 30 m LiDAR horizon yields 150 ms of warning and the stopping distance is roughly 2,000 m — no onboard sensor reaches it, so the problem is intercept geometry rather than avoidance. Ten milliseconds of latency is two meters of own-ship travel, or four meters of displacement against a head-on closure of 400 m/s. A pipeline that closes in 80 ms is inside the engagement envelope; one that closes in 1.4 s does not intercept.

Companion document. Every efficiency figure cited here is derived in “Measured Energy Efficiency of Embedded AI Compute,” which documents the 360-configuration measurement campaign, its methodology and its limitations. Every workload figure — op counts, byte counts, update rates, latency allocations — is derived line by line in the “Autonomy Workload — Data Annex.” This document is the argument; that one is the audit trail. branes.ai